

LIS 4370 R Markdown Primer

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Introduction

R Markdown is a way to combine text, math, and R code in one file. It helps keep analysis and writing together.

Narrative Paragraph

This document shows a basic workflow with short text and simple code. The output is easy to read because results appear under each code chunk. The example uses a simple single-cell RNA tumor dataset.

Math Examples

Inline math example: $\alpha + \beta = \gamma$.

Displayed equation:

$$y = mx + b$$

Basic R Code Chunk 1

```
sc_tumor <- data.frame(  
  cell_id = paste0("Cell", 1:8),  
  sample_type = c("Tumor", "Tumor", "Tumor", "Tumor", "Normal", "Normal", "Normal", "Normal"),  
  geneA_expr = c(8.2, 7.9, 9.1, 8.7, 3.2, 2.8, 3.5, 2.9)  
)  
  
head(sc_tumor)
```

```
##   cell_id sample_type geneA_expr  
## 1   Cell1         Tumor         8.2  
## 2   Cell2         Tumor         7.9  
## 3   Cell3         Tumor         9.1  
## 4   Cell4         Tumor         8.7  
## 5   Cell5         Normal         3.2  
## 6   Cell6         Normal         2.8
```

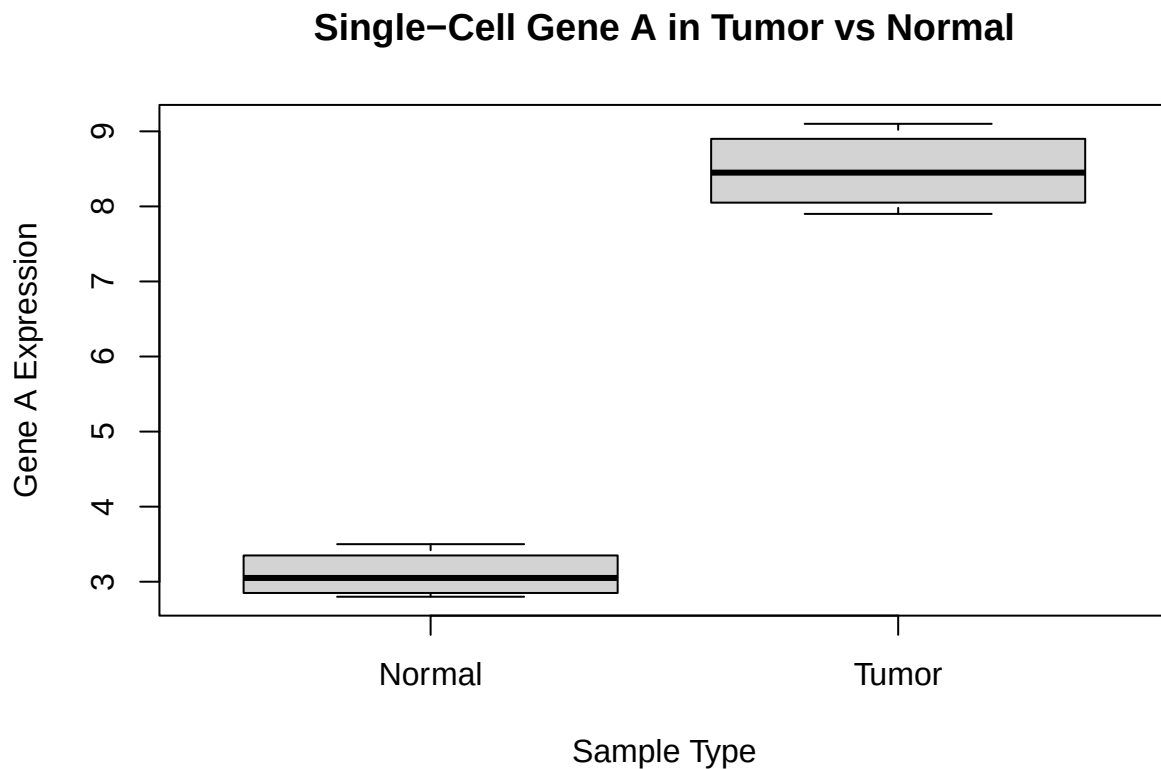
Basic R Code Chunk 2

In this box plot, the box measures the interquartile range, $IQR = Q_3 - Q_1$, which represents the middle 50% of Gene A expression values.

```
summary(sc_tumor$geneA_expr)
```

```
##      Min. 1st Qu.  Median    Mean 3rd Qu.    Max.
##      2.800   3.125   5.700   5.787   8.325   9.100
```

```
boxplot(geneA_expr ~ sample_type, data = sc_tumor,
        xlab = "Sample Type",
        ylab = "Gene A Expression",
        main = "Single-Cell Gene A in Tumor vs Normal")
```



Reflection

R Markdown feels easier than writing a plain report because code and output stay in one place. It also supports reproducible analysis since the same file can be rerun to regenerate results.

Compile and Review

When I compare source and output, I notice that headings and math are formatted more cleanly in the rendered file. One useful tip is to keep section names simple and keep code chunks short so errors are easier to find.